THE VIRTUAL MEET EXPERIENCE

2025-2026

HS VIRTUAL CHALLENGE MEET #5 - THE RQ MEET

COMPUTER SCIENCE

DO NOT OPEN TEST UNTIL TOLD TO DO SO

The Virtual Challenge Meets™

2025-2026 High School
Virtual Challenge Meet #5 - The RQ Meet

General Directions:

1) DO NOT OPEN EXAM UNTIL TOLD TO DO SO.
2) **NO CALCULATORS of any kind may be used.**
3) You have 45 minutes to complete this contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
4) Papers may not be turned in until forty-five minutes have elapsed. If you finish the test before the end of the allotted time, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
5) All answers must be written on the answer sheet/Scantron card provided. Indicate your answers in the appropriate blanks provided on the answer sheet or on the Scantron card. Clean erasures are necessary for accurate Scantron grading.
6) You may place as many notations as you desire anywhere on the test paper except on the answer sheet or Scantron card which is reserved for answers only.
7) You may use additional scratch paper provided by the contest director.
8) All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers. **All provided code segments are intended to be syntactically correct, unless otherwise stated (i.e. `error` is an answer choice). Ignore any typographical errors and assume any undefined variables are defined as used.**
9) A reference to commonly used Java classes is provided with the test and you may use this reference during the contest. You may detach the reference sheets from the test booklet but DO NOT DO SO UNTIL THE CONTEST BEGINS.
10) Assume that any necessary import statements for **Java 23** and classes (e.g. .lang, .util, System, Math, Double, etc.) are included in any programs or code segments that refer to methods from these classes and/or packages.

Scoring:

1) All questions will receive 6 points if answered correctly; no points will be given or subtracted if unanswered; 2 points will be deducted for each incorrect answer.
2)

For Computer Science practice tests and hands-on materials, go to www.apluscompsci.com

Standard Classes and Interfaces — Supplemental Reference

class java.lang.Object

- `boolean equals(Object other)`
- `String toString()`
- `int hashCode()`

interface java.lang.Comparable<T>

- `int compareTo(T other)`
  Return value < 0 if `this` is less than `other`.
  Return value = 0 if `this` is equal to `other`.
  Return value > 0 if `this` is greater than `other`.

class java.lang.Integer implements Comparable<Integer>

- `Integer(int value)`
- `int intValue()`
- `boolean equals(Object obj)`
- `String toString()`
- `int compareTo(Integer anotherInteger)`
- `static int parseInt(String s)`

class java.lang.Double implements Comparable<Double>

- `Double(double value)`
- `double doubleValue()`
- `boolean equals(Object obj)`
- `String toString()`
- `int compareTo(Double anotherDouble)`
- `static double parseDouble(String s)`

class java.lang.String implements Comparable<String>

- `int compareTo(String anotherString)`
- `boolean equals(Object obj)`
- `int length()`
- `String substring(int begin, int end)`
  Returns the substring starting at index `begin` and ending at index `(end - 1)`.
- `String substring(int begin)`
  Returns `substring(from, length())`.
- `int indexOf(String str)`
  Returns the index within this string of the first occurrence of `str`. Returns `-1` if `str` is not found.
- `int indexOf(String str, int fromIndex)`
  Returns the index within this string of the first occurrence of `str`, starting the search at the specified index.. Returns `-1` if `str` is not found.
- `charAt(int index)`
- `int indexOf(int ch)`
- `int indexOf(int ch, int fromIndex)`
- `String toLowerCase()`
- `String toUpperCase()`
- `String[] split(String regex)`
- `boolean matches(String regex)`

class java.lang.Character

- `static boolean isDigit(char ch)`
- `static boolean isLetter(char ch)`
- `static boolean isLetterOrDigit(char ch)`
- `static boolean isLowerCase(char ch)`
- `static boolean isUpperCase(char ch)`
- `static char toUpperCase(char ch)`
- `static char toLowerCase(char ch)`

class java.lang.Math

- `static int abs(int a)`
- `static double abs(double a)`
- `static double pow(double base, double exponent)`
- `static double sqrt(double a)`
- `static double ceil(double a)`
- `static double floor(double a)`
- `static double min(double a, double b)`
- `static double max(double a, double b)`
- `static int min(int a, in b)`
- `static int max(int a, int b)`
- `static long round(double a)`
- `static double random()`
  Returns a `double` value with a positive sign, greater than or equal to `0.0` and less than `1.0`.

interface java.util.List<E>

- `boolean add(E e)`
- `int size()`
- `Iterator<E> iterator()`
- `ListIterator<E> listIterator()`
- `E get(int index)`
- `E set(int index, E e)`
  Replaces the element at `index` with the object `e`.
- `void add(int index, E e)`
  Inserts the object `e` at position `index`, sliding elements at position `index` and higher to the right (adds 1 to their indices) and adjusts `size`.
- `E remove(int index)`
  Removes element from position `index`, sliding elements at position `(index + 1)` and higher to the left (subtracts 1 from their indices) and adjusts `size`.

class java.util.ArrayList<E> implements List<E>

class java.util.LinkedList<E> implements List<E>, Queue<E>

Methods in addition to the `List` methods:

- `void addFirst(E e)`
- `void addLast(E e)`
- `E getFirst()`
- `E getLast()`
- `E removeFirst()`
- `E removeLast()`

class java.util.Stack<E>
- `boolean isEmpty()`
- `E peek()`
- `E pop()`
- `E push(E item)`

interface java.util.Queue<E>
- `boolean add(E e)`
- `boolean isEmpty()`
- `E peek()`
- `E remove()`

class java.util.PriorityQueue<E>
- `boolean add(E e)`
- `boolean isEmpty()`
- `E peek()`
- `E remove()`

interface java.util.Set<E>
- `boolean add(E e)`
- `boolean contains(Object obj)`
- `boolean remove(Object obj)`
- `int size()`
- `Iterator<E> iterator()`
- `boolean addAll(Collection<? extends E> c)`
- `boolean removeAll(Collection<?> c)`
- `boolean retainAll(Collection<?> c)`

class java.util.HashSet<E> implements Set<E>

class java.util.TreeSet<E> implements Set<E>

interface java.util.Map<K,V>
- `Object put(K key, V value)`
- `V get(Object key)`
- `boolean containsKey(Object key)`
- `int size()`
- `Set<K> keySet()`
- `Set<Map.Entry<K, V>> entrySet()`

class java.util.HashMap<K,V> implements Map<K,V>

class java.util.TreeMap<K,V> implements Map<K,V>

interface java.util.Map.Entry<K,V>
- `K getKey()`
- `V getValue()`
- `V setValue(V value)`

interface java.util.Iterator<E>
- `boolean hasNext()`
- `E next()`
- `void remove()`

interface java.util.ListIterator<E> extends java.util.Iterator<E>

Methods in addition to the `Iterator` methods:
- `void add(E e)`
- `void set(E e)`

class java.lang.Exception
- `Exception()`
- `Exception(String message)`

class java.util.Scanner
- `Scanner(InputStream source)`
- `boolean hasNext()`
- `boolean hasNextInt()`
- `boolean hasNextDouble()`
- `String next()`
- `int nextInt()`
- `double nextDouble()`
- `String nextLine()`
- `Scanner useDelimiter(String pattern)`

2025-2026 Virtual Challenge Meet #5 – Computer Science

Note: Correct responses are based on **Java 23**. All provided code segments are intended to be syntactically correct, unless otherwise stated (i. e. `error` is an answer choice) and any necessary Java Standard Packages have been imported. Ignore any typographical errors and assume any undefined variables are defined as used. **For all output statements, assume that the System class has been statically imported using:** ***import static java.lang.System.*;***

QUESTION 1

What is 343_5 plus 1010010_2 ?

A. 265_8 B. bd_{16} C. 10010010_2 D. 2310_4 E. none of these

QUESTION 2

What is output by the code to the right?

A. 12 B. 17 C. 13 D. 20

E. There is no output due to a compile error.

```
out.println( 10 + 5 / 2 + 1 );
```

QUESTION 3

What is output by the code to the right?

A. ZZ3*2 B. ZZ32

C. ZZ6 D. ZZ

E. 6

```
out.println( "ZZ" + 3 * 2 );
```

QUESTION 4

What is output by the code to the right?

A. 22 B. 30

C. 25 D. 33

E. 28

```
String s = "www.apluscompsci.com";
s += "www.apluscompsci.com";
s = s.replaceAll("www.","@");
s = s.replaceAll("compsci","@");
out.println(s.length());
```

QUESTION 5

What is output by the code to the right?

A. true B. false

```
boolean z = true || false;
z = !z && false ^ true;
out.println( z );
```

QUESTION 6

What is output by the code to the right?

A. 3.0 B. 4.0

C. 3.5 D. 3

E. 4

```
double w = Math.ceil(3.1);
w = Math.round(w);
out.println( (int)w );
```

QUESTION 7

What is output by the code to the right?

A. 10 B. 8

C. 11 D. 9

E. 7

```
int b = 10;
int e = b / 3;
int q = e * 3;
out.println( q );
```

QUESTION 8

What is output by the code to the right?

A. 7 B. 57
C. 37 D. 35
E. 357

```
int u = 10;
if( u > 5 )
   out.print( 3 );
if( u > 7 )
   out.print( 5 );
out.print( 7 );
```

QUESTION 9

What is output by the code to the right?

A. 02468 B. 0246
C. 024 D. 2468
E. 0246810

```
for( int i=0; i<10; i+=2 )
   out.print(i);
```

QUESTION 10

What is output by the code to the right ?

A. 31 B. 24
C. 30 D. 27
E. 22

```
int[] x = {2,4,6,8,10,15,20};
out.print( x[1] + x[3] + x[5] );
```

QUESTION 11

What is output by the code to the right?

A. 10 B. -12
C. -9 D. 9
E. 7

```
String s = "";
s = "practice.apluscompsci.com";
s = "2 1 -5 6 -9 3 8";
Scanner z = new Scanner(s);
int x = z.nextInt();
z.nextInt();
x += z.nextInt();
z.nextInt();
x += z.nextInt();
out.println( x );
```

QUESTION 12

What is output by the code to the right?

A. 37 B. 43
C. 40 D. 35
E. 46

```
int k = 0;
for( int i=0; i<20; i+=4 )
   k += i;
out.print(k);
```

QUESTION 13

What is output by the code to the right ?

A. 27 B. 30
C. 32 D. 23
E. 31

```
out.print( 5 << 1 | 20 ^ 5 + 2 );
```

QUESTION 14

What is output by the code to the right?

A. -127 B. 127
C. 256 D. 128
E. -128

```
int z = Byte.MAX_VALUE;
z += 1;
out.print( z );
```

QUESTION 15

What is output by the code to the right ?

A. `[Java.util.ArrayList]`

B. `[(this Collection)]`

C. unable to be determined until runtime

D. there is no output due to a syntax error in the code

E. there is no output due to a runtime error

```
ArrayList<ArrayList>list;
list = new ArrayList<ArrayList>();
list.add(list);
out.println(list);
```

QUESTION 16

What is output by the code to the right?

A. `UIL` B. `APLUS`

C. `CS` D. `BEST`

E. `FUN`

```
String s = "APLUSUILBESTUILCSUILFUN";
String[] r = s.split("UIL");
out.print( r[2] );
```

QUESTION 17

What is output by the code to the right?

A. `001 011 101 110`

B. `000 010 100 111`

C. `010 011 101 110`

D. `001 101 101 110`

E. `001 011 101 111`

```
for(int i = 0;i<=1;i++)
   for(int j = 0;j<=1;j++)
      out.print(i+""+j+(i&j)+" ");
```

QUESTION 18

What is output by the code to the right ?

A. `-1` B. `4`

C. `7` D. `9`

E. There is no output due to a runtime error.

```
String s = "";
s = "practice.apluscompsci.com";
s += "aplusaplusaplus";
out.print( s.indexOf( "aplus" ) );
```

QUESTION 19

What is returned by the method call `aplus(42)` ?

A. `101010` B. `10101`

C. `110110` D. `111001`

E. `111000`

QUESTION 20

What is returned by the method call `aplus(0xBAD)` ?

A. `101101011101` B. `1011010101`

C. `110100100101` D. `10000101010`

E. `101110101101`

```
public static String aplus(int i)
{
   if(i==0)
     return "";
   return aplus(i/2)+i%2;
}
```

QUESTION 21

What is output by `//line 1` in the code to the right?

A. 1
B. 2
C. 3
D. 4
E. 6

QUESTION 22

What is output by `//line 2` in the code to the right?

A. 1
B. 2
C. 3
D. 6
E. 9

QUESTION 23

What is output by `//line 3` in the code to the right?

A. 1
B. 3
C. 4
D. 6
E. 12

```
class A
{
  public int x = 1;
  public void setX(int i)
  {
     x = i;
  }
  public void fun()
  {
     go();
  }
  public void go()
  {
     setX(x*2);
  }
}
class B extends A{
  public int x = 3;
  public void go()
  {
     super.go();
     setX(x*2);
  }
}

//////////////////////////////////////
//client code
A a = new A();
a.fun();
out.println(a.x);  //line 1
A b = new B();
out.println(b.x);  //line 2
b.fun();
out.println(b.x);  //line 3
```

QUESTION 24

What is output by `//line 1` in the code to the right?

A. 1
B. 6
C. null
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

QUESTION 25

What is output by `//line 2` in the code to the right?

A. 1
B. 6
C. null
D. There is no output due to a compile error.
E. There is no output due to a runtime error.

```
Map<Integer,Integer> m;
m = new HashMap<>();

m.put(3,6);
m.put(8,2);
m.put(12,10);
m.put(9,212);

Object t = m.putIfAbsent(3,1);
out.println(t);  //line 1

out.println(m.get(3));  //line 2
```

QUESTION 26

What is the returned by the call `go("dodges","dogeb")`?

A. 1 B. 2 C. 3 D. 4 E. 5

QUESTION 27

What is returned by the call

`go("APLUS COMP SCI","APCSIH")`?

A. 1 B. 3 C. 5 D. 7 E. 9

```
public int go(String a, String b){
  if(a.isEmpty()||b.isEmpty())
     return 0;
  String suba = a.substring(1);
  String subb = b.substring(1);
  if(a.charAt(0)==(b.charAt(0)))
     return 1+go(suba,subb);
  int x = go(suba,b);
  int y = go(a,subb);
  return Math.max(x,y);
}
```

QUESTION 28

What is output by `//line 1` in the code to the right?

A. 2
B. 5
C. -3
D. 7
E. 4

QUESTION 29

What is output by `//line 2` in the code to the right?

A. 2
B. 5
C. -3
D. 7
E. 1

QUESTION 30

What is output by `//line 3` in the code to the right?

A. 2
B. 8
C. -3
D. 7
E. 4

```
ArrayList<Integer> r;
r = new ArrayList<>();
r.add(2);
r.add(5);
r.add(8);
r.add(0,4);
r.add(1);
r.add(3,7);
r.add(-3);

out.println( r.get(2) );  //line 1


r.remove(2);
r.remove(3);

out.println( r.get(3) );   //line 2

Collections.sort( r );
Collections.rotate( r, -2 );

out.println( r.get(2) );   //line 3
```

QUESTION 31

What is output by the code to the right ?

A. true B. false
C. 7.0 D. 13.0
E. 17.0

```
out.println(Math.hypot(5,12));
```

QUESTION 32

What is output by `//line 1` in the code to the right?

A. `1`

B. `2`

C. `3`

D. `-1`

E. `0`

QUESTION 33

What is output by `//line 2` in the code to the right?

A. `true`

B. `false`

C. `null`

D. There is no output due to a compile error.

E. There is no output due to a runtime error.

```
Stack<Integer> s;
s = new Stack<>();
Stack<Integer> t;
t = new Stack<>();
s.add(3);
s.push(2);
s.add(11);
s.push(5);
s.pop();
s.add(8);
s.pop();
s.add(7);

out.println(s.indexOf(2) ); //line 1

t=s;
for( int x : s )
   t.add(x);

out.println(s.equals(t)); //line 2
```

QUESTION 34

What is output by `//line 1` in the code to the right?

A. `0` B. `1`

C. `2` D. `3`

E. `4`

QUESTION 35

What is output by `//line 2` in the code to the right?

A. `-1` B. `0`

C. `1` D. `2`

E. there is no output due to a runtime exception

```
LinkedList<Integer> list;
list = new LinkedList<Integer>();
for(int i = 0;i<10;i++)
list.add(i);
ListIterator<Integer> it;
it = list.listIterator();
it.next();
it.next();
out.println(it.next());//line 1
it.remove();
it.add(-1);
it.add(-1);
out.println(it.previous());//line 2
```

QUESTION 36

What is output by the code to the right ?

A. `50` B. `2`

C. `51` D. `3`

E. There is no output due to a compile error.

```
int z = '0';
z++;
char c = (char)z;
c++;
z = c;
z++;
c = (char)z;
out.println( c );
```

QUESTION 37

Which data structure does class structure represent?

A. a set

B. a map

C. a linked list

D. a min heap

E. a binary search tree

QUESTION 38

There is one underlying fundamental problem with this data structure that gives rise to many more. What is the problem ?

A. removing from an empty structure causes an exception

B. you cannot add to the structure once it is full

C. some indices can be overwritten if two Object's haschode wrap to the same position

D. remove may not find the right value

E. remove will not work if there is only 1 item

F. the code works fine as it is

```
class structure
{
  Object[] d;

  public structure(int cap){
     d = new Object[cap];
     Arrays.fill(d,-1);
  }

  public void add(Object o,Object p)
  {
     d[o.hashCode()%d.length] = p;
  }

  public Object find(Object o)
  {
     return d[o.hashCode()%d.length];
  }

  public void remove(Object o)
  {
     d[o.hashCode()] = -1;
  }
}
```

QUESTION 39

Name one of the primary graph algorithms that can handle negative weighted edges.

QUESTION 40

What type of binary tree is one where every node has either zero or two children?

Name _____

School _____

Division: 1A 2A 3A 4A 5A 6A Grade: 9 10 11 12

Final Score _____ _____

2nd Grading _____ _____

1st Grading _____ _____
score initials

The Virtual Challenge Meets
Computer Science - Student Answer Sheet

1) _____
2) _____
3) _____
4) _____
5) _____
6) _____
7) _____
8) _____
9) _____
10) _____

11) _____
12) _____
13) _____
14) _____
15) _____
16) _____
17) _____
18) _____
19) _____
20) _____

21) _____
22) _____
23) _____
24) _____
25) _____
26) _____
27) _____
28) _____
29) _____
30) _____

31) _____
32) _____
33) _____
34) _____
35) _____
36) _____
37) _____
38) _____
39) _____
40) _____

Total Number Correct _____ x 6 = _____

Total Number Incorrect _____ x -2 = _____

Final Score (record here and top right) _____

2025-2026
Virtual Challenge Meet #5 - The RQ Meet
Computer Science - KEY

1) D
2) C
3) C
4) A
5) B
6) E
7) D
8) E
9) A
10) D

11) B
12) C
13) A
14) D
15) B
16) C
17) B
18) D
19) A
20) E

21) B
22) A
23) D
24) B
25) B
26) D
27) C
28) B
29) E
30) D

31) D
32) A
33) E
34) C
35) A
36) D
37) B
38) C
39) Bellman-Ford or Floyd-Warshall
40) full binary tree

Note to Graders:

- All provided code segments are intended to be syntactically correct, unless otherwise stated (e.g. error is an answer). **Ignore any typographical errors**.
- Any necessary Standard Java 23 Packages are assumed to have been imported as needed.
- Assume any undefined (undeclared) variables have been defined as used.

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